Global Economics

Grade 10 – Term 1 – Classroom activity 2

A decision under uncertainty selects an alternative A_i before a state S_j is known. Each pair (A_i, S_j) determines a payoff

$$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j),$$

placed in row A_i, column S_j. The objective is to construct the complete payoff table.

Revenue and cost are

$$R = q(A_i, S_j)\big[r(A_i) + r(S_j)\big] + f_r(A_i) + f_r(S_j),$$

$$C = q(A_i, S_j)\big[c(A_i) + c(S_j)\big] + f_c(A_i) + f_c(S_j),$$

$$P(A_i, S_j) = q(A_i, S_j)\big[r(A_i) + r(S_j) - c(A_i) - c(S_j)\big] + f_r(A_i) + f_r(S_j) - f_c(A_i) - f_c(S_j).$$

Variable components are per-unit rates multiplied by $q(A_i, S_j)$; fixed components are not. A_i, S_j, and (A_i, S_j) indicate dependence on thealternative, state, or both. Missing components are zero; directly given R or C require no decomposition.

Context Descriptions

For each problem, **(C1)** *identify the decision alternatives, the states of nature, the uncertain event, and the consequence.* Then **(C3)** *calculate revenue, cost, and profit for every alternative–state pair and construct the payoff table.*

1. **Direct case.** A small café must decide whether to launch an artisan-dessert line or keep the current menu before foot traffic is known. Traffic may be high or low. All values are in thousand USD. Launching the dessert line produces revenue and cost of 58 and 16 under high traffic, and 27 and 19 under low traffic. Keeping the current menu produces revenue and cost of 49 and 21 under high traffic, and 41 and 21 under low traffic.
2. **One-component problem.** A custom-printing cooperative must choose between producing premium notebooks and producing canvas bags before market conditions are known. Conditions may be strong demand with stable material prices or weak demand with higher material prices. The respective quantities are 6 and 4 production lots for premium notebooks, and 5 and 3 lots for canvas bags. Revenue per lot depends on the product: 12 thousand USD for premium notebooks and 15 thousand USD for canvas bags. Cost per lot depends on the state: 5 thousand USD under strong demand and stable prices, and 8 thousand USD under weak demand and higher prices.
3. **Two-component problem.** A farm cooperative must choose between a standard produce box and a premium produce box before seasonal demand is known. Demand may be strong or weak. Quantities, in hundreds of subscriptions, are 7 and 4 for the standard box and 6 and 3 for the premium box under strong and weak demand, respectively. The plan-based revenue rates are 8 and 11 thousand USD per hundred subscriptions for the standard and premium boxes. Strong demand adds a state-based revenue rate of 3 thousand USD per hundred subscriptions; weak demand adds 1. The plan-based cost rates are 4 and 6 thousand USD per hundred subscriptions, respectively. The fixed distribution costs are 9 thousand USD under strong demand and 13 thousand USD under weak demand.
4. **Three-component problem.** A regional logistics company must choose a standard or express delivery plan before demand and traffic conditions are known. Conditions may be high demand with light traffic or moderate demand with heavy traffic. Quantities, in hundreds of delivery batches, are 8 and 5 for the standard plan and 10 and 6 for the express plan under the two states, respectively. The plan-based revenue rates are 9 and 11 thousand USD per hundred batches; the state-based revenue rates are 3 and 2; and the plan-based fixed revenues are 16 and 24 thousand USD. The plan-based cost rates are 4 and 5 thousand USD per hundred batches; the state-based cost rates are 1 and 3; and the state-based fixed costs are 10 and 15 thousand USD.

1 Solutions: Revenue, Costs, Profit, and Payoff Tables

Problems:

1.1. Local Café Investment (C1–C3)

1.2. Custom Printing Product Line (C1–C3)

1.3. Farm Produce Subscription Plans (C1–C3)

1.4. Regional Parcel Delivery (C1–C3)

1.1 Exercise 1: Local café investment (direct case)

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Launch dessert line	A_2: Keep current menu
States of nature	S_1: High traffic	S_2: Low traffic
Uncertain event	Actual foot traffic after the menu decision	
Consequences	Operating the selected activity under the realised conditions leaves funds available or a shortfall to cover. The monetary measure is profit in thousand USD for each alternative–state pair.	



C3 - Calculating profit and building the payoff table.

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Launch dessert line; A_2: Keep current menu	Alternative labels, in row order.
S_j	S_1: High traffic; S_2: Low traffic	State labels, in column order; revealed after the choice.
$R(A_i, S_j)$	$(1,1)$: 58 thousand USD; $(1,2)$: 27 thousand USD; $(2,1)$: 49 thousand USD; $(2,2)$: 41 thousand USD	Supplied total revenue for each indexed pair; no component breakdown.
$C(A_i, S_j)$	$(1,1)$: 16 thousand USD; $(1,2)$: 19 thousand USD; $(2,1)$: 21 thousand USD; $(2,2)$: 21 thousand USD	Supplied total cost for each indexed pair; no component breakdown.

The total revenue and total cost are supplied directly for every cell, so

$$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j).$$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1,S_1) - C(A_1,S_1) = P(A_1,S_1)$	$R(A_1,S_2) - C(A_1,S_2) = P(A_1,S_2)$
A_2	$R(A_2,S_1) - C(A_2,S_1) = P(A_2,S_1)$	$R(A_2,S_2) - C(A_2,S_2) = P(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$58-16 = 42$	$27-19 = 8$
A_2	$49-21 = 28$	$41-21 = 20$

Final payoff table (profit in thousand USD).

Alternative	High traffic (S_1)	Low traffic (S_2)
Launch dessert line (A_1)	42	8
Keep current menu (A_2)	28	20

1.2 Exercise 2: Custom printing product line (one component)

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Produce premium notebooks	A_2: Produce canvas bags
States of nature	S_1: Strong demand and stable material prices	S_2: Weak demand and higher material prices
Uncertain event	Market conditions revealed after the product-line decision	
Consequences	Operating the selected activity under the realised conditions leaves funds available or a shortfall to cover. The monetary measure is profit in thousand USD for each product-line–market-state pair.	



C3 - Calculating profit and building the payoff table.

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Produce premium notebooks; A_2: Produce canvas bags	Alternative labels, in row order.
S_j	S_1: Strong demand and stable material prices; S_2: Weak demand and higher material prices	State labels, in column order; revealed after the choice.
$q(A_i, S_j)$	$(1,1)$: 6 production lots; $(1,2)$: 4 production lots; $(2,1)$: 5 production lots; $(2,2)$: 3 production lots	Quantity under each alternative and state.
$r(A_i)$	12; 15 thousand USD/lot	Revenue per lot for premium notebooks; canvas bags. Values follow the listed index order.
$c(S_j)$	5; 8 thousand USD/lot	Cost per lot under strong/stable; weak/higher-price conditions. Values follow the listed index order.

Revenue has one alternative-based variable component, while cost has one state-based variable component:

$$R = q(A_i, S_j)r(A_i), \qquad C = q(A_i, S_j)c(S_j), \qquad P = R - C.$$

Revenue

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1, S_1) \times r(A_1)$ $= R(A_1, S_1)$	$q(A_1, S_2) \times r(A_1)$ $= R(A_1, S_2)$
A_2	$q(A_2, S_1) \times r(A_2)$ $= R(A_2, S_1)$	$q(A_2, S_2) \times r(A_2)$ $= R(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	6×12 $= 72$	4×12 $= 48$
A_2	5×15 $= 75$	3×15 $= 45$

Cost

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1, S_1) \times c(S_1)$ $= C(A_1, S_1)$	$q(A_1, S_2) \times c(S_2)$ $= C(A_1, S_2)$
A_2	$q(A_2, S_1) \times c(S_1)$ $= C(A_2, S_1)$	$q(A_2, S_2) \times c(S_2)$ $= C(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	6×5 $= 30$	4×8 $= 32$
A_2	5×5 $= 25$	3×8 $= 24$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1, S_1) - C(A_1, S_1)$ $= P(A_1, S_1)$	$R(A_1, S_2) - C(A_1, S_2)$ $= P(A_1, S_2)$
A_2	$R(A_2, S_1) - C(A_2, S_1)$ $= P(A_2, S_1)$	$R(A_2, S_2) - C(A_2, S_2)$ $= P(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$72 - 30$ $= 42$	$48 - 32$ $= 16$
A_2	$75 - 25$ $= 50$	$45 - 24$ $= 21$

Final payoff table (profit in thousand USD).

Alternative	Strong demand, stable prices (S_1)	Weak demand, higher prices (S_2)
Premium notebooks (A_1)	42	16
Canvas bags (A_2)	50	21

1.3 Exercise 3: Farm produce subscription plans (two components)

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Offer a standard produce box	A_2: Offer a premium produce box
States of nature	S_1: Strong seasonal demand	S_2: Weak seasonal demand
Uncertain event	Seasonal market conditions revealed after the subscription-plan decision	
Consequences	Operating the selected activity under the realised conditions leaves funds available or a shortfall to cover. The monetary measure is profit in thousand USD for each plan–market-state pair.	



C3 - Calculating profit and building the payoff table.

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Offer a standard produce box; A_2: Offer a premium produce box	Alternative labels, in row order.
S_j	S_1: Strong seasonal demand; S_2: Weak seasonal demand	State labels, in column order; revealed after the choice.
$q(A_i, S_j)$	$(1,1)$: 7 hundreds of subscriptions; $(1,2)$: 4 hundreds of subscriptions; $(2,1)$: 6 hundreds of subscriptions; $(2,2)$: 3 hundreds of subscriptions	Quantity under each alternative and state.
$r(A_i)$	8; 11 thousand USD/100 subscriptions	Revenue rate for standard; premium boxes. Values follow the listed index order.
$r(S_j)$	3; 1 thousand USD/100 subscriptions	Extra revenue rate under strong; weak demand. Values follow the listed index order.
$c(A_i)$	4; 6 thousand USD/100 subscriptions	Cost rate for standard; premium boxes. Values follow the listed index order.
$f_c(S_j)$	9; 13 thousand USD	Fixed distribution cost under strong; weak demand. Values follow the listed index order.

Revenue combines alternative-based and state-based variable components. Cost combines an alternative-based variable component and a state-based fixed component:

$$R = q(A_i, S_j)[r(A_i) + r(S_j)], \quad C = q(A_i, S_j)c(A_i) + f_c(S_j), \quad P = R - C.$$

Revenue

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1, S_1)\times [r(A_1)+r(S_1)] = R(A_1, S_1)$	$q(A_1, S_2)\times [r(A_1)+r(S_2)] = R(A_1, S_2)$
A_2	$q(A_2, S_1)\times [r(A_2)+r(S_1)] = R(A_2, S_1)$	$q(A_2, S_2)\times [r(A_2)+r(S_2)] = R(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$7\times [8+3] = 77$	$4\times [8+1] = 36$
A_2	$6\times [11+3] = 84$	$3\times [11+1] = 36$

Cost

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1, S_1)\times c(A_1) + f_c(S_1) = C(A_1, S_1)$	$q(A_1, S_2)\times c(A_1) + f_c(S_2) = C(A_1, S_2)$
A_2	$q(A_2, S_1)\times c(A_2) + f_c(S_1) = C(A_2, S_1)$	$q(A_2, S_2)\times c(A_2) + f_c(S_2) = C(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$7\times 4 + 9 = 37$	$4\times 4 + 13 = 29$
A_2	$6\times 6 + 9 = 45$	$3\times 6 + 13 = 31$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1, S_1) - C(A_1, S_1) = P(A_1, S_1)$	$R(A_1, S_2) - C(A_1, S_2) = P(A_1, S_2)$
A_2	$R(A_2, S_1) - C(A_2, S_1) = P(A_2, S_1)$	$R(A_2, S_2) - C(A_2, S_2) = P(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$77 - 37 = 40$	$36 - 29 = 7$
A_2	$84 - 45 = 39$	$36 - 31 = 5$

Final payoff table (profit in thousand USD).

Alternative	Strong seasonal demand (S_1)	Weak seasonal demand (S_2)
Standard produce box (A_1)	40	7
Premium produce box (A_2)	39	5

1.4 Exercise 4: Regional parcel delivery (three components)

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Use the standard delivery plan	A_2: Use the express delivery plan
States of nature	S_1: High demand with light traffic	S_2: Moderate demand with heavy traffic
Uncertain event	Demand and traffic conditions revealed after the delivery-plan decision	
Consequences	Operating the selected activity under the realised conditions leaves funds available or a shortfall to cover. The monetary measure is profit in thousand USD for each delivery-plan–state pair.	



C3 - Calculating profit and building the payoff table.

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Use the standard delivery plan; A_2: Use the express delivery plan	Alternative labels, in row order.
S_j	S_1: High demand with light traffic; S_2: Moderate demand with heavy traffic	State labels, in column order; revealed after the choice.
$q(A_i, S_j)$	$(1,1)$: 8 hundreds of delivery batches; $(1,2)$: 5 hundreds of delivery batches; $(2,1)$: 10 hundreds of delivery batches; $(2,2)$: 6 hundreds of delivery batches	Quantity under each alternative and state.
$r(A_i)$	9; 11 thousand USD/100 batches	Revenue rate for standard; express plans. Values follow the listed index order.
$r(S_j)$	3; 2 thousand USD/100 batches	Extra revenue rate under light; heavy traffic. Values follow the listed index order.
$f_r(A_i)$	16; 24 thousand USD	Fixed revenue for standard; express plans. Values follow the listed index order.
$c(A_i)$	4; 5 thousand USD/100 batches	Cost rate for standard; express plans. Values follow the listed index order.
$c(S_j)$	1; 3 thousand USD/100 batches	Extra cost rate under light; heavy traffic. Values follow the listed index order.
$f_c(S_j)$	10; 15 thousand USD	Fixed cost under light; heavy traffic. Values follow the listed index order.

Revenue has alternative- and state-based variable components and an alternative-based fixed component. Cost has alternative- and state-based variable components and a state-based fixed component:

$$R = q(A_i, S_j)[r(A_i) + r(S_j)] + f_r(A_i),$$
$$C = q(A_i, S_j)[c(A_i) + c(S_j)] + f_c(S_j),$$
$$P = R - C.$$

Revenue

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times$ $[r(A_1)+r(S_1)]$ $+f_r(A_1)$ $=R(A_1,S_1)$	$q(A_1,S_2)\times$ $[r(A_1)+r(S_2)]$ $+f_r(A_1)$ $=R(A_1,S_2)$
A_2	$q(A_2,S_1)\times$ $[r(A_2)+r(S_1)]$ $+f_r(A_2)$ $=R(A_2,S_1)$	$q(A_2,S_2)\times$ $[r(A_2)+r(S_2)]$ $+f_r(A_2)$ $=R(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$8\times$ $[9+3]$ $+16$ $=112$	$5\times$ $[9+2]$ $+16$ $=71$
A_2	$10\times$ $[11+3]$ $+24$ $=164$	$6\times$ $[11+2]$ $+24$ $=102$

Cost

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times$ $[c(A_1)+c(S_1)]$ $+f_c(S_1)$ $=C(A_1,S_1)$	$q(A_1,S_2)\times$ $[c(A_1)+c(S_2)]$ $+f_c(S_2)$ $=C(A_1,S_2)$
A_2	$q(A_2,S_1)\times$ $[c(A_2)+c(S_1)]$ $+f_c(S_1)$ $=C(A_2,S_1)$	$q(A_2,S_2)\times$ $[c(A_2)+c(S_2)]$ $+f_c(S_2)$ $=C(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$8\times$ $[4+1]$ $+10$ $=50$	$5\times$ $[4+3]$ $+15$ $=50$
A_2	$10\times$ $[5+1]$ $+10$ $=70$	$6\times$ $[5+3]$ $+15$ $=63$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1,S_1)-C(A_1,S_1)$ $=P(A_1,S_1)$	$R(A_1,S_2)-C(A_1,S_2)$ $=P(A_1,S_2)$
A_2	$R(A_2,S_1)-C(A_2,S_1)$ $=P(A_2,S_1)$	$R(A_2,S_2)-C(A_2,S_2)$ $=P(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$112-50$ $=62$	$71-50$ $=21$
A_2	$164-70$ $=94$	$102-63$ $=39$

Final payoff table (profit in thousand USD).

Alternative	High demand, light traffic (S_1)	Moderate demand, heavy traffic (S_2)
Standard delivery plan (A_1)	62	21
Express delivery plan (A_2)	94	39